

Open source, openness and standardization of large models

Hang Chen

China Electronics Standardization Institute (CESI)



CONTENTS

GOSIM Paris 2026

Part 01

CESI: Who are we

Part 02

Mulan OSC: What have we done

Part 03

AI-OpenG: Open source, openness and standardization of large models



01

CESI: Who are we



1、CESI: Who are we

Established in 1963, the China Electronics Standardization Institute (CESI) is a public institution directly under the Ministry of Industry and Information Technology (MIIT).

With electronic information technology standardization at its core, CESI conducts scientific research, testing, metrology, certification, and information services based on standards.

CESI has established partnerships with international standardization organizations and foreign institutions, playing a significant role in promoting the application of standards.

45

Responsible for 45 domestic mirror organizations for ISO、IEC、ISO/IEC JTC1

17

Responsible for 17 National Technical Committees on Standardization

24

Responsible for 24 subcommittees of the National Technical Committee on Standardization



Office infrastructure in Beijing, Shanghai, Shenzhen...



02 Mulan OSC: What have we done



MulanPSL-2.0 is the only international open source license recognized by OSI, and officially available in both Chinese and English

MulanPSL-2.0

- ❑ Grant of Copyright License;
- ❑ Grant of Patent License;
- ❑ No Trademark License;
- ❑ The Chinese version and the English version shall have the same legal effect;
- ❑ More than 100,000 open-source projects have adopted the MulanPSL-2.0;



- ❑ Link address:

<http://opensource.org/licenses/MulanPSL-2.0>

<http://www.apache.org/legal/resolved.html#category-a>



2、Mulan OSC、CCF Ubiquitous OS Open Community

GOSIM Paris 2026

We have initiated and operated the Mulan OSC、CCF Ubiquitous OS Open Community, dedicated to promoting the development of open-source technologies in fields such as AI, operating systems, and basic software.



中国光华科技基金会
CHINA GUANGHUA FOUNDATION

Collaborated with the China Guanghua Science and Technology Foundation to establish an open-source operation capability and provide development funds for open-source projects.

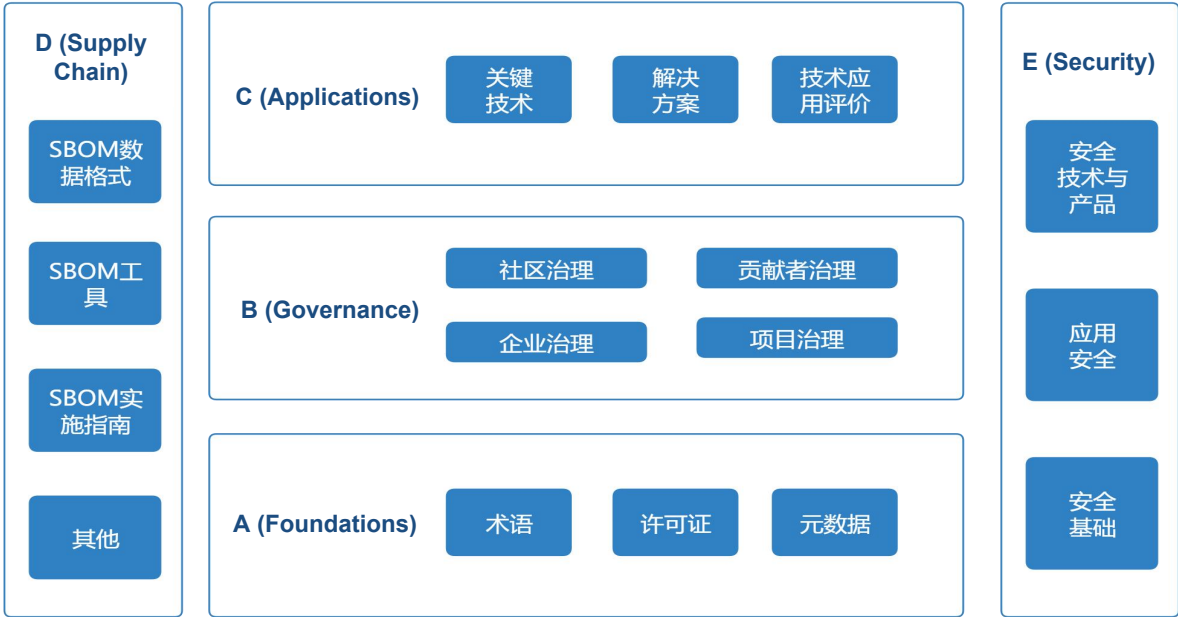


Launched open-source innovation competition for developers worldwide, incubated open-source projects such as industrial IoT operating system.

Five platinum-tier members and more than thirty member organizations



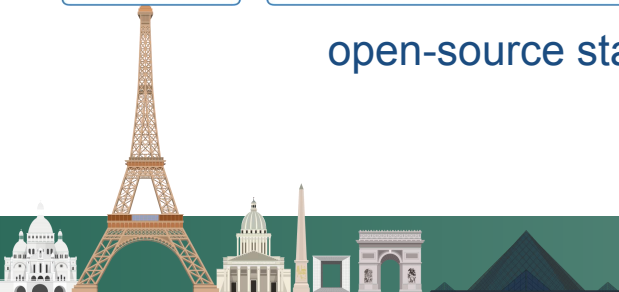
Taking into account the key directions and critical aspects of the open-source ecosystem, we have established an open-source standards framework comprising six components: A (Foundations), B (Governance), C (Applications), D (Supply Chain), and E (Security).



open-source standards framework

No.	Category	Standard No./Plan No.	Standard Title
1	A (Foundations)	GB/T 44272-2024	Open-Source License Framework
2		T/CESA 1269-2023	Open Source Terminology and Overview
3		T/CESA 1291-2023	General Requirements for Open-Source Metadata
4	B (Governance)	T/CESA 1270.4-2023	Open-Source Project Evaluation Model
5		T/CESA 1270.5-2023	Open Source Contributor Evaluation Model
6			
7			
8	C (Applications)	2025-0257T-SJ	Specifications for the Graded Evaluation of Open-Source and Open-Access Capabilities in Large-Scale Model Technology Systems
9			
10	D (Supply Chain)	2024-1400T-SJ	Specifications for Software Bill of Materials Data Formats
11			
12			
13	E (Security)	GB/T43698—2024	Software Supply Chain Security Requirements
14			
15			

List of Open Source Standards (Partial)



3、Open Source Ecosystem Events

GOSIM Paris 2026

We have been organizing ecosystem events such as the “Mulan Technology Open Day”, China Open Source Hackathon and so on, to accelerate innovation and adoption of open-source technologies.



Mulan Technology Open Day



Mulan Open Source Conference



China Open Source Hackathon



Open-Source Programming Summer Camp



ISO/IEC International Standards Meeting

...



03

Open Source, Openness and Standardization of Large Models

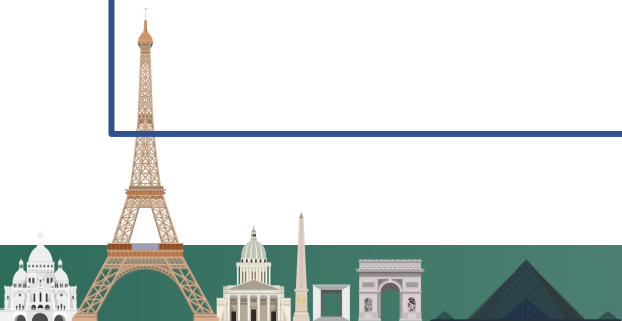


Core Issues

- Existing international frameworks are **incompatible with the realities of domestic development**. (such as ISO's OSAID, MOF from the Linux Foundation, etc.)
- Mainstream open-source large models (such as Llama, DeepSeek, etc.) have **ill-defined openness boundaries, with frequent licensing disputes**.

Standard Benefits

- **Clarify hierarchical boundaries** of large model open access, and **provide scientific guidance** for developers to optimize model materials.
- **Mitigate large model security** and compliance risks, and **support users' model selection decision-making**.
- **Stimulate technological innovation**, and foster a sound, sustainable open-source ecosystem.



2、Five-Level Grading Framework for Open-Source and Openness of Large Models

This standard establishes a **five-level grading framework** for the open-source openness of large model technology systems based on two core dimensions: **completeness and openness**.

- **Completeness:** Reflects the level of **disclosure of technical resources** related to large models, as well as the **degree of support for tasks such as model reproduction and validation**. It progresses incrementally from L1 to L5, with **higher levels encompassing all requirements of lower levels**.
- **Openness:** Reflects the **authorization boundaries set by the license** and **the agreement for users to use, modify and redistribute the model** and its related resources. Under the same completeness level, openness is divided into four categories from O1 to O4, which ascends as the burden of licensing clauses decreases.

		开放性 O（许可自由度）			
		O1 （有限制）	O2 （互惠）	O3 （宽松）	O4 （零义务）
完整性 L （资源完备度）	L1 开放部署级	L1O1	L1O2	L1O3	L1O4
	L2 模型理解级	L2O1	L2O2	L2O3	L2O4
	L3 训练公开级	L3O1	L3O2	L3O3	L3O4
	L4 模型验证级	L4O1	L4O2	L4O3	L4O4
	L5 全链开源级	L5O1	L5O2	L5O3	L5O4



3、Grading Requirements for Open Source and Open Access of Large Models

Level	Completeness	Openness			
		Code License	Data License	Model License	
				Sub-category	Restrictive Clauses
L1	Baseline items: Model weights; Model configuration files; Tokenizer files; Inference scripts; Inference environment instructions;	Open Source Software License	Specify if sample data is attached; otherwise, mark as "—".	L101	Prohibited from use in the improvement of other models
					Mandatory prominent brand display
					Commercial use prohibited
					Commercial use restricted
					Geographic or national scope restrictions
				L102	Derivatives shall be open-sourced under identical or compatible licenses
				L103	No substantial usage restrictions; closed-source retention or license replacement upon redistribution is allowed
L2	New additions: Model principle description; Training process explanation; Evaluation results;	If training-related code is disclosed, an open source software license is required; otherwise, mark as "—".	Specify if training-related samples/metadata are disclosed; otherwise, mark as "—".	L104	Fully unrestricted for free use
				L201	Restrictive clauses identical to L101
				L202	Derivatives shall be open-sourced under identical or compatible licenses
				L203	No substantial usage restrictions; closed-source retention or license replacement upon redistribution is allowed
L3	New additions: Training code; Training data; Training environment instructions;	Open Source Software License	The license shall be clearly defined and authorize licensees to conduct model training.	L204	Fully unrestricted for free use
				L301	Restrictive clauses identical to L101
				L302	Derivatives shall be open-sourced under identical or compatible licenses
				L303	No substantial usage restrictions; closed-source retention or license replacement upon redistribution is allowed
L4	New additions: Evaluation code; Evaluation data; Evaluation environment instructions;	Open Source Software License	The license shall be clearly defined and authorize licensees to conduct model testing.	L304	/
				L401	Restrictive clauses identical to L101
				L402	Original license terms shall be inherited and publicly disclosed
				L403	No substantial usage restrictions; closed-source retention or license replacement upon redistribution is allowed
L5	New additions: Data preprocessing scripts; Data cleaning scripts; Raw data description;	Open Source Software License	Specify if raw data samples/metadata are disclosed; otherwise, mark as "—".	L404	/
				L501	Restrictive clauses identical to L101
				L502	Original license terms shall be inherited and publicly disclosed
				L503	No substantial usage restrictions; closed-source retention or license replacement upon redistribution is allowed

4、 Grading Example – L1

Level	Release Completeness Requirements	Code License	Data License	Model License		Description	Representative Model
				Subcategory	Restrictive Clauses		
L1	Baseline items: Model weights; Model configuration files; Tokenizer files; Inference scripts; Inference environment instructions;	Open Source Software License	Specify if sample data is attached; otherwise marked as "—"	L1O1	Prohibited for the enhancement of other models	It is prohibited to use the model's materials, outputs or derivatives to improve other models.	xai-org/grok-2

Model card

Files and versions

Community

main grok-2

Go to file

3 contributors

History: 14 commits

LilyGrok

Update LICENSE

d60cbe2

VERIFIED

4 days ago

.gitattributes

Safe

1.52 kB

initial commit

6 days ago

LICENSE

Safe

5.36 kB

Update LICENSE

4 days ago

README.md

Safe

1.58 kB

Update README.md (#1)

5 days ago

config.json

Safe

947 Bytes

Update config.json

5 days ago

pytorch_model-00000-TP-common.safetensors

Safe

2.15 GB

Add files using upload-large-folder tool

5 days ago

pytorch_model-00001-TP-common.safetensors

Safe

2.15 GB

Add files using upload-large-folder tool

5 days ago

pytorch_model-00002-TP-common.safetensors

Safe

16.5 GB

Add files using upload-large-folder tool

5 days ago

pytorch_model-00003-TP-common.safetensors

Safe

34.4 GB

Add files using upload-large-folder tool

5 days ago

pytorch_model-00004-TP-common.safetensors

Safe

34.4 GB

Add files using upload-large-folder tool

5 days ago

pytorch_model-00005-TP-common.safetensors

Safe

34.4 GB

Add files using upload-large-folder tool

5 days ago

pytorch_model-00006-TP-000.safetensors

Safe

17.2 GB

Add files using upload-large-folder tool

5 days ago

pytorch_model-00006-TP-001.safetensors

Safe

17.2 GB

Add files using upload-large-folder tool

5 days ago

pytorch_model-00006-TP-002.safetensors

Safe

17.2 GB

Add files using upload-large-folder tool

5 days ago

pytorch_model-00017-TP-common.safetensors

Safe

8.4 MB

Add files using upload-large-folder tool

5 days ago

tokenizer.json

Safe

7.72 MB

Add files using upload-large-folder tool

5 days ago



Grok 2

This repository contains the weights of Grok 2, a model trained and used at xAI in 2024.

Usage: Serving with SGLang

Download the weights. You can replace /local/grok-2 with any other folder name you prefer.

hf download xai-org/grok-2 --local-dir /local/grok-2

You might encounter some errors during the download. Please retry until the download is successful.

If the download succeeds, the folder should contain 42 files and be approximately 500 GB.

Launch a server.

Install the latest SGLang inference engine (>= v0.5.1) from https://github.com/sgl-project/sglang/

Use the command below to launch an inference server. This checkpoint is TP=8, so you will need 8 GPUs (each with > 40GB of memory).

python3 -m sglang.launch_server --model /local/grok-2 --tokenizer-path /local/

Send a request.

This is a post-trained model, so please use the correct chat template.

python3 -m sglang.test_send_one --prompt "Human: What is your name?
Assistant:"

You should be able to see the model output its name, Grok.

Learn more about other ways to send requests here.

License

The weights are licensed under the Grok 2 Community License Agreement.

Grok 2 Community License Agreement

Last Updated: August 23, 2025

1. Background and Definitions

By downloading, accessing, or using the Materials (as defined below) related to Grok 2 provided by xAI ("xAI"), you ("Licensee" or "you") agree to the terms of this agreement.

2. License Grant & Scope

a. Permitted Uses: xAI grants you a non-exclusive, worldwide, revocable license to use, reproduce, distribute, and modify the Materials:

For non-commercial and research purposes; and

For commercial use solely if you and your affiliates abide by all of the guardrails provided in xAI's Acceptable Use Policy (https://x.ai/legal/acceptable-use-policy).

b. Restrictions:

You may not use the Materials, derivatives, or outputs (including generated data) to train, create, or improve any foundational, large language, or general-purpose model.

No right to use xAI's trademarks is granted, except as required for attribution (see below).

3. Distribution & Attribution

If you distribute the Materials, derivatives, or products/services incorporating them:

Include this Agreement and a notice stating: "This product includes materials licensed under the xAI Community License. Copyright © 2025 xAI. All rights reserved."

Prominently display "Powered by xAI" in related materials or interfaces.

4. Ownership & Outputs

xAI retains all rights in the Materials. This Agreement does not impose any restrictions or obligations with respect to any use, modification, or sharing of any outputs generated by the Materials.

5. Acceptable Use

You are responsible for implementing appropriate safety measures, including filters and human oversight, suitable for your use case. You must comply with xAI's Acceptable Use Policy.

6. Warranty Disclaimer & Limitation of Liability

THE MATERIALS ARE PROVIDED "AS IS", WITHOUT WARRANTY OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO THE WARRANTIES OF MERCHANTABILITY, FITNESS FOR A PARTICULAR PURPOSE, OR NON-INFRINGEMENT. YOU ARE SOLELY RESPONSIBLE FOR (1) CLEARING RIGHTS OF OTHER PERSONS THAT MAY APPLY TO THE MATERIALS OR ANY USE THEREOF, INCLUDING WITHOUT LIMITATION ANY PERSON'S COPYRIGHTS IN NO EVENT SHALL XAI BE LIABLE FOR ANY CLAIM, DAMAGES, OR OTHER LIABILITY, WHETHER IN AN ACTION OF CONTRACT, TORT, OR OTHERWISE, ARISING FROM, OUT OF, OR IN CONNECTION WITH THE MATERIALS.

7. Termination

This license terminates immediately upon your breach or if you exceed the permitted commercial threshold. Upon termination, you must cease all use and delete all copies of the Materials.

8. Governing Law

The laws of Texas govern this Agreement, and any dispute shall be resolved exclusively in the courts located in Tarrant County, Texas.

9. Miscellaneous

This Agreement is the entire agreement between the parties on this subject. Failure to enforce any provision is not a waiver. If any provision is unenforceable, the remaining provisions shall survive.

Level	Release Completeness Requirements	Code License	Data License	Model License		Description	Representative Model
				Subcategory	Restrictive Clauses		
L2	New additions: Model principle description; Training process explanation; Evaluation results;	If training-related code is disclosed, an open source software license is required; otherwise, mark as "—".	Specify if training-related samples or metadata are disclosed; otherwise, indicate with "—".	L2O1	Restrictive clauses identical to L1O1	May be closed-sourced or re-licensed upon redistribution, but prohibited for high-risk and harmful use cases.	deepseek-ai/DeepSeek-V3

DeepSeek-V3 Technical Report

DeepSeek-AI, Aixin Liu, Bei Feng, Bing Xue, Bingxuan Wang, Bochao Wu, Chengda Lu, Cheng Dongjie Ji, Erhang Li, Fangyun Lin, Fucong Dai, Fuli Luo, Guangbo Hao, Guanting Chen, Guow Gao, Hui Li, Hui Qu, J.L. Cai, Jian Liang, Jianzhong Guo, Jiaqi Ni, Jiashi Li, Jiawei Wang, Jin Ci Kang Guan, Kexin Huang, Kuai Yu, Lean Wang, Lecong Zhang, Lei Xu, Leyi Xia, Liang Zhao, Li Mingming Li, Ning Tian, Panpan Huang, Peiyi Wang, Peng Zhang, Qiancheng Wang, Qihao Zhu, Ruoyu Zhang, Ruyi Chen, S.S. Li, Shanghao Lu, Shangyan Zhou, Shanhuang Chen, Shaoqing Shuting Pan, T. Wang, Tao Yun, Tian Pei, Tianyu Sun, W.L. Xiao, Wangding Zeng et al. (100 ad

We present DeepSeek-V3, a strong Mixture-of-Experts (MoE) language model with 671B total parameters with Latent Attention (MLA) and DeepSeekMoE architectures, which were thoroughly validated in DeepSeek-V2. For training objective for stronger performance. We pre-train DeepSeek-V3 on 14.8 trillion diverse and high-quality Comprehensive evaluations reveal that DeepSeek-V3 outperforms other open-source models and achieves 2.788M H800 GPU hours for its full training. In addition, its training process is remarkably stable. Throughout checkpoints are available at [this https URL](#).

Subjects: **Computation and Language (cs.CL)**; Artificial Intelligence (cs.AI)
Cite as: [arXiv:2412.19437 \[cs.CL\]](#)
(or [arXiv:2412.19437v2 \[cs.CL\]](#) for this version)
<https://doi.org/10.48550/arXiv.2412.19437> ⓘ

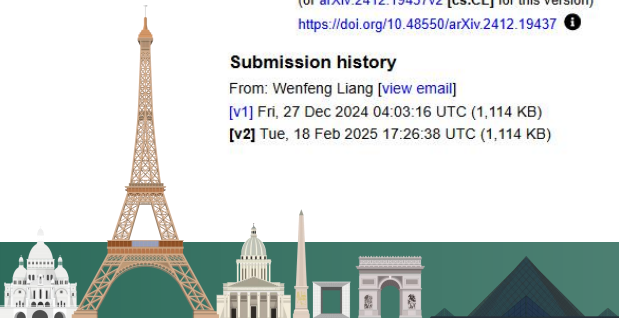
Submission history

From: Wenfeng Liang [view email]
[v1] Fri, 27 Dec 2024 04:03:16 UTC (1,114 KB)
[v2] Tue, 18 Feb 2025 17:26:38 UTC (1,114 KB)

Model principle description;
Training process explanation

Category	Benchmark (Metric)	Claude-3.5-Sonnet-1022	GPT-4o-0513	DeepSeek V3	OpenAI o1-mini	OpenAI o1-1217	DeepSeek R1
English	Architecture	-	-	MoE	-	-	MoE
	# Activated Params	-	-	37B	-	-	37B
	# Total Params	-	-	671B	-	-	671B
	MMLU (Pass@1)	88.3	87.2	88.5	85.2	91.8	90.8
	MMLU-Redux (EM)	88.9	88.0	89.1	86.7	-	92.9
	MMLU-Pro (EM)	78.0	72.6	75.9	80.3	-	84.0
	DROP (3-shot F1)	88.3	83.7	91.6	83.9	90.2	92.2

Evaluation results



Level	Release Completeness Requirements	Code License	Data License	Model License		Description	Representative Model
				Subcategory	Restrictive Clauses		
L3	New additions: Training code; Training data; Training environment instructions;	Open Source Software License	The license shall be clearly defined and authorize licensees to conduct model training.	L3O2	Derivatives shall be open-sourced under identical or compatible licenses	When redistributing training code, configurations or derived weights, the original license must be inherited and any modifications shall be publicly disclosed.	multimolecule/rnafm

 ml4bio / RNA-FM



Code

Issues 15

Pull requests

Actions

Projects

Security

Insights

 RNA-FM

Public

Watch 6

Fork 32

Star 298

main

1 Branch

0 Tags

Go to file

Add file

Code

 WangJiuming

Update pretrained.py

3489515 · 3 months ago

47 Commits

docs	# This is a combination of 23 commits.	4 months ago
fm	Update pretrained.py	3 months ago
redevelop	mrna-fm	last year
tutorials	# This is a combination of 23 commits.	4 months ago

Nature Methods: RNA foundation model (together with RhoFold)

ml4bio.github.io/RNA-FM/

Readme

MIT license

Activity

Custom properties

Training Details

RNA-FM used Masked Language Modeling (MLM) as the pre-training objective: taking a sequence, the model randomly masks 15% of the tokens in the input then runs the entire masked sentence through the model and has to predict the masked tokens. This is comparable to the Cloze task in language modeling.

Training data

Training Data

The RNA-FM model was pre-trained on [RNAcentral](#). RNAcentral is a free, public resource that offers integrated access to a comprehensive and up-to-date set of non-coding RNA sequences provided by a collaborating group of [Expert Databases](#) representing a broad range of organisms and RNA types.

RNA-FM applied [CD-HIT \(CD-HIT-EST\)](#) with a cut-off at 100% sequence identity to remove redundancy from the RNAcentral. The final dataset contains 23.7 million non-redundant RNA sequences.

RNA-FM preprocessed all tokens by replacing "U"s with "T"s.

Note that during model conversions, "T" is replaced with "U". [RnaTokenizer] [multimolecule.RnaTokenizer] will convert "T"s to "U"s for you, you may disable this behaviour by passing `replace_T_with_U=False`.

Training code、 Training environment instructions



7、 Grading Example – L4

Level	Release Completeness Requirements	Code License	Data License	Model License		Description	Representative Model
				Subcategory	Restrictive Clauses		
L4	New additions: Evaluation code; Evaluation data; Evaluation environment instructions;	Open Source Software License	The license shall be clearly defined and authorize licensees to conduct model testing.	L4O3	/	Capable of verifying the authenticity of model performance, and permitting free use of licensed or derivative materials based on the model.	mosaicml/mpt-7b

 mosaicml / llm-foundry

Type to search

Code

Issues 40

Pull requests 17

Actions

Projects

Security

Insights

 llm-foundry

Public

Watch 50

Fork 578

Star 4.3k

main

336 Branches

32 Tags

Go to file

Add file

Code

 rithwik-db

Gating megablocks based on CUDA availability (#1865) ✓

1bc4dd0 · last month

1,349 Commits

 .github

Change test to use foundry images (#1839)

3 months ago

 .pre-commit

Remove .ci folder and move FILE_HEADER (#931)

last year

 assets

Model gauntlet (#308)

2 years ago

About

LLM training code for Databricks foundation models

www.databricks.com/blog/introducing-d...

nlp

deep-learning

pytorch

neural-networks

llm

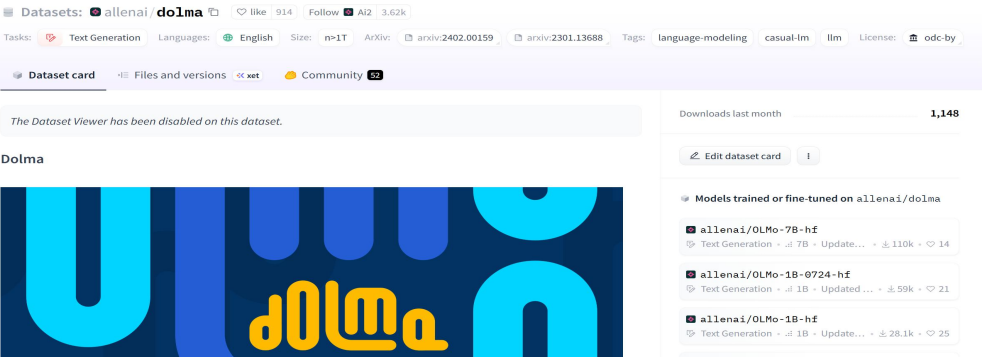
Readme

Evaluation code;
Evaluation data;
Evaluation environment instructions;

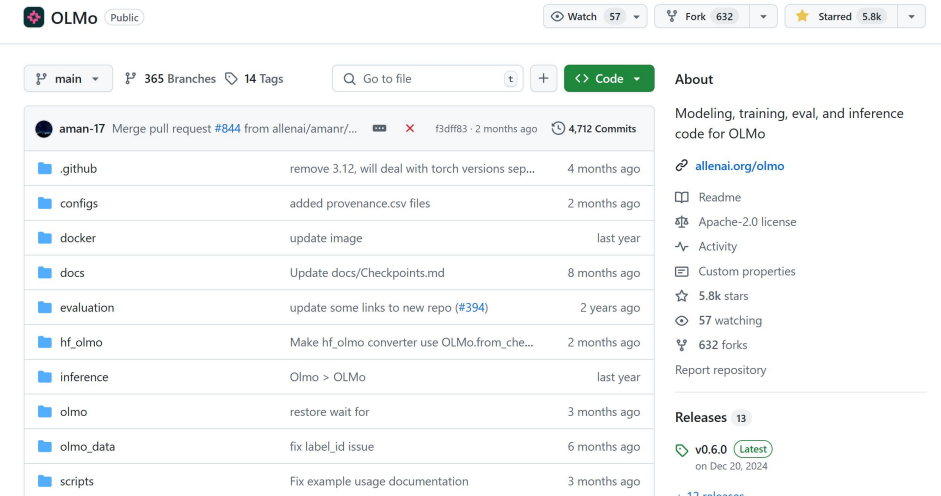


8、 Grading Example – L5

Level	Release Completeness Requirements	Code License	Data License	Model License		Description	Representative Model
				Subcategory	Restrictive Clauses		
L5	New additions: Data preprocessing scripts; Data cleaning scripts; Raw data description; Reproduction guidelines;	Open Source Software License	Specify if raw data samples/metadata are disclosed; otherwise, mark as "—".	L5O3	/	Meets the requirements for full-process reproducibility of training, evaluation and deployment, as well as methodological interpretability; licensed and derivative materials based on the model may be used without restriction.	allenai/OLMo-7B



Raw data description



Data preprocessing scripts;
Data cleaning scripts;
Reproduction guidelines



Standardized Evaluation



- **Evaluation Basis:** 《Artificial intelligence — Key fundamental technologies — Large-scale model technology architecture open source and openness level evaluation specification》
- **Evaluation Method:** **Tool-based testing** + **manual assessment**
- **Evaluation Content:** **Complete open-source materials** of the large model provided by the tested party, including model weight files, configuration files, license compliance documents, etc.

Value



- **For model tested parties:** **Improve the completeness** of model materials and meet compliance requirements.
- **For model users:** Support **accurate model selection**, and **avoid security and compliance risks** while matching business needs.



Thanks!

